

BILAL AHAMAD

Lead Embedded Firmware & Systems QA Engineer

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SUMMARY

Systems validation architect with 18+ years building embedded software validation ecosystems for safety-critical systems — firmware, IoT and wearables, mobile handsets, AI video pipelines, and autonomous driving. Takes products from NPI bring-up to enterprise-scale release: test automation, Software-in-the-Loop environments, test strategy, and lab infrastructure that cuts risk and accelerates feedback.

CORE SKILLS

Languages & Automation: Python, Shell, C/C++, JavaScript, Pytest, Selenium, Appium

OS & Embedded Hardware: Embedded Linux, RTOS, QNX, Android/AAOS, iOS, ADB, ARM, ESP32, Raspberry Pi, Arduino

Wireless, Protocols & Diagnostics: BLE, LTE/Wi-Fi, Wi-Fi Alliance (WFA) certification, QXDM, RTSP/RTP, TLS, UART/SPI/I2C, CAN/CAN FD, Automotive Ethernet, V2X, Wireshark/PCAP replay

Infrastructure & Observability: CI/CD (GitLab CI, GitHub Actions, Buildkite, Jenkins), SIL (Docker/QEMU), HIL benches, Bazel, Git, JIRA, Splunk, Grafana, Datadog, Databricks

EXPERIENCE

System Architect & Technical QA Lead — [Stealth Mode](#) Sep 2025 - Aug 2026 · Remote / Sunnyvale, CA
Led architecture and validation for a sensor-integration developer IDE, from a LiDAR/Radar bring-up platform through to a Unified Sensor Test Platform.

- Built a LiDAR/Radar bring-up platform in Python with Docker/QEMU SIL (virtual-ECU) environments and a pre-validated ARM/ESP32 compatibility layer, cutting bring-up cycles by 70%.
- Defined technical strategy for the Unified Sensor Test Platform, carrying it from framework concept to production-ready release gates with containerized A/B testing and benchmarking.

Senior Firmware Quality Lead — [Samsara Inc](#) Dec 2023 - Jul 2025 · Remote / San Francisco, CA
Owned firmware quality for commercial-fleet IoT Dash Cams in an AI/ML video safety ecosystem, leading 4 engineers across direct and matrixed teams.

- Cut the regression cycle from two weeks to five days and post-launch firmware escapes by 30% across the Dash Cam product line.
- Wrote Pytest automation across the Edge-to-Cloud capture-encode-upload pipeline on Embedded Linux, covering on-device AI precision, security, and network robustness (RTSP/RTP, TLS, LTE/Wi-Fi).

Senior Automation Architect — [Cruise LLC](#) Oct 2022 - Jun 2023 · San Francisco, CA
Charted the test-delivery roadmap for safety-critical vehicle computational systems, defining ASIL-D validation protocols with systems architects for compute-platform releases.

- Engineered a Python framework stress-testing autonomous driving compute workloads across CPU/GPU integration, AI/ML performance, V2X and PCAP replay, reaching 75% automation coverage.
- Qualified CPU/GPU compute platforms and firmware releases on HIL benches against performance, reliability, safety, fault-tolerance, and redundancy benchmarks.

Test Lead & Senior System Test Engineer — [Rivian Automotive LLC](#) Jun 2021 - Sep 2022 · Palo Alto, CA
Directed test strategy and system test quality ownership across R1T, R1S, and Commercial Fleet (EDV): multi-ECU handshaking, bootloader validation, and low-power/12V body control.

- Reduced production-blocking navigation and connectivity issues by 40% before first customer ship, coordinating 3+ squads on a bi-weekly release train.
- Architected the Python automation framework for QNX and Android covering OTA updates, navigation, infotainment and UI/UX, running 24/7 on SIL/HIL benches through Buildkite CI.

Senior Product Quality Lead & QAE II — Amazon Lab126

Jun 2018 - Jun 2021 · Sunnyvale, CA

Led the quality roadmap for Alexa Voice Service products (Echo Buds, Echo Auto, Dash Button) across 50+ localized SKUs, synchronizing DSP, Acoustic and Firmware teams.

- Engineered a reusable automation library of AVS test primitives, reducing manual testing cost by \$3.0M.
- Ran 24/7 CI/CD pipelines with firmware quality gates and live Splunk dashboards across the device test labs, cutting post-launch field issues by 30%.

Senior Test Engineer — Tech Mahindra / Google Inc

Jan 2016 - Jun 2018 · Mountain View, CA

Steered system-level validation and firmware release sign-off across Pixel 2, Pixel 3 and the Google Daydream VR ecosystem, coordinating readiness with 10+ stakeholders.

- Designed and built a 3DOF robotic motion-stimulus fixture (Arduino, servo motors) automating motion-fidelity, IMU sensor-fusion and spatial-tracking validation, cutting manual execution hours by 80%.
- Developed Python suites for UART bus analysis and real-time battery profiling across charge/discharge cycling, Depth of Discharge (DoD), and hibernation states.

Product Validation Engineer — Cognizant Technology / Cisco

Sep 2015 - Jan 2016 · Boston, MA

- Cut execution cycles by 50% while increasing repeatability, expanding test scope by 40% through modular architecture and programmable actions.

Radio Validation Engineer — Wistron Mobile Solutions Corp

Dec 2014 - Sep 2015 · Rolling Meadows, IL

- Automated the WLAN and interoperability (IOT) test harnesses to standardize handset certification validation, delivering \$1.3M in operational savings.

Senior Software Engineer & Test Automation Lead — L&T Infotech / Motorola Mobility

Oct 2009 - Dec 2014 · Chicago, IL

- Architected a 3-tier Bluetooth qualification framework, enhancing code reusability 3x and reducing test preparation LOE by 70%.

Software Developer — Luminous Infoways

Oct 2008 - Sep 2009 · Bhubaneswar, India

- Delivered web application deployment modules and feature integration, and served as the client liaison on portal releases.

SELECTED PROJECTS

US Live Layoff Monitoring Dashboard — A national WARN layoff dataset unifying notices from 46 states and DC into one searchable feed with a free JSON API and per-state email alerts — refreshed twice daily by a GitHub Actions pipeline with ETag caching and MD5 verification, zero human intervention.

Adhan Caster — Cross-Browser Extension — A Manifest V3 browser extension that auto-pauses every playing video and audio across all open tabs at Adhan time, with a live countdown to the next prayer. One codebase live on the Chrome Web Store, Firefox Add-ons and Edge Add-ons — built in one day with Claude Code.

Smart-Home IoT Media Caster — An IoT orchestration layer that casts automated prayer-time notifications to a Sony Android TV, built in 4 days instead of a 3-week engineering cycle with AI pair programming. Integrates a Raspberry Pi over ADB, managing media state and low-level device commands.

CERTIFICATIONS

- Google AI Professional (2026)
- Google Project Management Professional (2026)
- ISTQB Certified Tester Foundation Level (CTFL)
- AI Coding Agents with GitHub Copilot and Cursor (2025)

EDUCATION

B.Tech, Electronics & Telecommunications Engineering
Biju Patnaik University of Technology —
2004 – 2008